33 kV FROM HV SWITCHBOARD
F00-SW-010
2000-F00-SLD-EL-10006

F23-SR-002
415V FINAL TAILS PUMPS SWITCHROOM

F23-TR-023
TRANSFORMER ROOM

F23-TR-023
33/0.433kV
1000kVA
Dyn11
z=6%
ANAF
NOTE 3

SWR EARTH BAR
Tx EARTH BAR
1600/ 1A
2.5VA
5P20

F23-SW-023
FINAL TAILS PUMPS 415V MOTOR CONTROL CENTRE

MCC EARTH BAR
NEUTRAL BAR
NOTE 1
1/700A
1600/ 5A
5VA
CL 1
00F-01
1600/ 5A
20VA
CL 0.5
00F-08
TEST
PM
20A
63A
2A
63A
240VAC SPRING
CHARGE SUPPLY
AFD
ACB
1600A
2A
24VDC CONTROL POWER SUPPLY
NOTE 7
100A
100A
SD
NOTE 6

415V 3NE~ 50Hz 1600A 65kA for 1sec

2R-11	3R-11	2F-01	3F-01	3F-05	3F-09	INC	1R-01		3F-13		
400A	400A	20A	20A	20A	20A		630A		160A		
2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A		2A 32A		2A 32A		
250A	250A	70A	70A	70A	70A		400A		70A		
EOL	EOL	EOL	EOL	EOL	EOL						
132kW	132kW	3.7kW	3.7kW	3.7kW	3.7kW						
F00-PS-464 EVENT POND SUBMERSIBLE PUMP 2000-F00-SCM-EL-10211	F00-PS-465 SILT TRAP SUBMERSIBLE PUMP 2000-F00-SCM-EL-10212	F23-FN-462 FINAL TAILS PUMP 1 MOTOR COOLING FAN 2000-F23-SCM-EL-10218	F23-FN-463 FINAL TAILS PUMP 2 MOTOR COOLING FAN 2000-F23-SCM-EL-10219	F23-FN-464 FINAL TAILS PUMP 3 MOTOR COOLING FAN 2000-F23-SCM-EL-10220	F23-FN-465 FINAL TAILS PUMP 4 MOTOR COOLING FAN 2000-F23-SCM-EL-10221						

160A
160A

FEEDER TO F23-SW-123
2000-F23-SCM-EL-10209

E-SHUT
F23SR002-LCS
NOTE 12

160A
F23-DB-101
FINAL TAILS PUMPS
11kV SWITCHROOM
HV VSD AUXILIARY SUUPLIES DB
2000-F23-SCM-EL-10224

CONTINUED ON SHT 2
2000-F23-SLD-EL-10202

NORTHERN STAR RESOURCES LIMITED

Approved	X
Approved as Noted	
Not Approved	
For Information Only	

Signed [redacted] Date 15/10/2025

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC AND ON TRANSFORMER READING "M.E.N. LINK LOCATED IN F23-SW-023 INCOMER TIER" CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE".
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. UPSTREAM TRANSFORMER FEEDER SHALL BE TRIPPED ON ACTIVATION OF SWITCHROOM EMERGENCY PUSHBUTTON, DETECTION OF EARTH FAULT OR ARC FAULT DETECTION.
4. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
5. ALL DRIVE I/O VIA COMMS NETWORK TO DCS. FOR COMMUNICATIONS BLOCK DIAGRAM REFER 2000-F23-BLK-IC-10001.
6. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
7. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
8. THERMAL OVERLOAD UNITS MAY BE SUBSTITUTED WITH ELECTRONIC (MAGNETIC) OVERLOAD UNITS TO SUIT DUTY.
9. TOLs TO BE MOUNTED IN PURPOSE-BUILT ENCLOSURE BENEATH VSD. TOLs MAY BE INCLUDED IN MCC MODULE WITH ENGINEERING APPROVAL.
10. ANY DRIVE RTDs SHALL BE WIRED TO DEDICATED REMOTE I/O CARDS IN THE FIELD. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
11. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
12. EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON N/O WIRED TO SHUNT TRIP NOMINATED CIRCUIT BREAKERS.

LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 4, 5
- DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 4, 5
- EOL — ELECTRONIC OVERLOAD UNIT NOTE 4, 5
- THERMAL OVERLOAD UNIT NOTE 8
- VARIABLE SPEED DRIVE NOTE 4, 5
- EARTH LEAKAGE RELAY NOTE 4, 5
- FIELD-MOUNTED DECONTACTOR
- MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
- RTD — MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
- MOTOR HEATERS (ONLY REQ. FOR MOTORS >90kW)
- 49 63 71 — TRANSFORMER PROTECTIVE DEVICES (REFER ANSI STANDARDS FOR DESCRIPTIONS)
- CIRCUIT BREAKER (NOTE 11)
- xxxA ST TRIP SET POINT / xxxA CB SIZE
- AFD — ARC FLASH DETECTION
- ECR — EARTH CONTINUITY RELAY
- 1 E-SHUT — SWITCHROOM EMERGENCY SHUTDOWN
- 63A 3 PHASE SOCKET OUTLET

DRG No.	REFERENCES
2000-F23-SCM-EL-10203	F23-SW-023 415V MCC ARC FLASH SYSTEM SCHEMATIC DIAGRAM
2000-F23-SCM-EL-10202	F23-SW-023 415V MCC 24VDC DISTRIBUTION SCHEMATIC DIAGRAM
2000-F23-SCM-EL-10201	F23-SW-023 415V MCC INCOMER SCHEMATIC DIAGRAM
2000-F23-DRG-EL-10001	F23-SR-002 415V SWITCHROOM ELECTRICAL EQUIPMENT LAYOUT
2000-F23-BLK-IC-10001	F23-SR-001 415V SWITCHROOM AND REMOTE I/O COMMUNICATIONS BLOCK DIAGRAM
2000-F23-SLD-EL-10202	F23-SW-023 415V MCC SINGLE LINE DIAGRAM SHEET 2
2000-F00-SLD-EL-10002	F00-SW-010 33kV SWITCHBOARD SINGLE LINE DIAGRAM SHEET 1 OF 5
2000-F00-STD-EL-10001	GENERAL NOTES AND STANDARD DRAWINGS INDEX

REV	DATE	DESCRIPTION	DRAW	CHECK	DESIGN	TECH APP	PROJ APP
1	19SEP25	FAT REDLINES INCORPORATED					
0	30MAY24	APPROVED FOR CONSTRUCTION					

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PROJECT: NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT

TITLE: FINAL TAILS HANDLING AND STORAGE
F23-SW-023 415V MCC
SINGLE LINE DIAGRAM SHEET 1

SHEET	SCALE	DRG No.	REVISION
A1	NONE	2000-F23-SLD-EL-10201	1

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\235ST KCGM GROWTH PROJECT\03 DELIVERABLES\EL\F23 - FINAL TAILINGS HANDLING & STORAGE\2000-F23-SLD-EL-10201.DWG 02OCT25 9:04AM HOTSEAT04